

Chapter 1.1 Probability Spaces

Rick Durrett Probability: Theory and Examples

MSU Probability Prelim

Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.1

Learning goal

Speak fluently about events, sigma-fields, probability measures, and extension from simple generating classes.

1. Probability spaces and sigma-fields.
2. Continuity of measure.
3. Discrete and Stieltjes probability models.
4. Extension from semialgebras and rectangles.
5. Exercise patterns: generators, increasing sigma-fields, density counterexamples.

Probability Space

Definition

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where Ω is the sample space, $\mathcal{F}$ is a sigma-field of events, and $\mathbb{P}$ is a countably additive measure with $\mathbb{P}(\Omega) = 1$.

- Events are only the subsets in $\mathcal{F}$.
- Most elementary probability identities come from closure of $\mathcal{F}$ and countable additivity of $\mathbb{P}$.

[LLMwiki: Probability space](#); ID: durrett_ch1_probability_space

Sigma-Field Closure Rules

Definition

A class $\mathcal{F}$ is a sigma-field if:

$$\Omega \in \mathcal{F}, \quad A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}, \quad A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Consequences

$\mathcal{F}$ is also closed under countable intersections, finite unions, finite intersections, and set differences.

[LLMwiki: Sigma-field closure rules](#); ID: durrett_ch1_sigma_field

How to Prove Sigma-Field Closure Consequences

Target

Derive countable intersection closure from complement and countable union closure.

Proof.

If $A_n \in \mathcal{F}$, then $A_n^c \in \mathcal{F}$ for every n . Since $\mathcal{F}$ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F}.$$

Taking complements and using De Morgan's law,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c \in \mathcal{F}.$$

Finite operations are special cases of countable operations. □

Basic Probability Identities

For $A, B \in \mathcal{F}$:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B),$$

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A) \quad (A \subseteq B),$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Teaching point

These are not separate axioms. They are quick consequences of disjoint decomposition and countable additivity.

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Continuity of Measure

Theorem

If $A_n \uparrow A$, meaning $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_n A_n$, then

$$\mathbb{P}(A_n) \uparrow \mathbb{P}(A).$$

If $A_n \downarrow A$, meaning $A_1 \supseteq A_2 \supseteq \dots$ and $A = \bigcap_n A_n$, then

$$\mathbb{P}(A_n) \downarrow \mathbb{P}(A).$$

- For general measures, the decreasing version needs $\mu(A_1) < \infty$.
- Under probability measures, this finiteness is automatic.

[LLMwiki: Continuity of measure](#); ID: durrett_ch1_measure_continuity

Proof Idea: Continuity from Below

Key move

Turn nested sets into disjoint increments.

Proof.

Let $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. Then the B_n are pairwise disjoint and

$$A_n = \bigcup_{k=1}^n B_k, \quad A = \bigcup_{k=1}^{\infty} B_k.$$

By countable additivity,

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$



Proof Idea: Continuity from Above

Key move

Reduce decreasing sets to increasing complements.

Proof.

If $A_n \downarrow A$, then $A_n^c \uparrow A^c$. By continuity from below,

$$\mathbb{P}(A_n^c) \uparrow \mathbb{P}(A^c).$$

Since $\mathbb{P}(A_n) = 1 - \mathbb{P}(A_n^c)$ and $\mathbb{P}(A) = 1 - \mathbb{P}(A^c)$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = 1 - \lim_{n \rightarrow \infty} \mathbb{P}(A_n^c) = 1 - \mathbb{P}(A^c) = \mathbb{P}(A).$$



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Discrete Probability Spaces

Model

If Ω is countable and $p(\omega) \geq 0$ with

$$\sum_{\omega \in \Omega} p(\omega) = 1,$$

then define

$$\mathbb{P}(A) = \sum_{\omega \in A} p(\omega), \quad A \subseteq \Omega.$$

- Countable additivity follows from rearranging nonnegative series.
- This intuition does not determine laws on an uncountable space.

[LLMwiki: Discrete probability spaces](#); ID: durrett_ch1_discrete_probability_space

Stieltjes Measures from Distribution-Like Functions

Theorem

A nondecreasing right-continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$ with suitable limits defines a unique Borel measure through interval increments:

$$\mu((a, b]) = F(b) - F(a).$$

If $F(-\infty) = 0$ and $F(\infty) = 1$, then μ is a probability measure.

Why it matters

This is the bridge from CDFs to probability measures on $\mathbb{R}$.

LLMwiki: [Stieltjes measures from distribution-like functions](#); ID: `durrett_ch1_stieltjes_measure`

How to Prove a Stieltjes Construction

1. Define $\mu((a, b]) = F(b) - F(a)$ on half-open intervals.
2. Verify nonnegativity and finite additivity on disjoint interval decompositions.
3. Use right-continuity of F to get the needed continuity property.
4. Extend from the interval semialgebra to $\mathcal{B}(\mathbb{R})$.
5. Use uniqueness because intervals generate the Borel sigma-field.

Instructor note

In an exam solution, one usually cites the extension theorem after checking that the interval formula is consistent and countably additive on the generating semialgebra. The proof burden is not to rebuild Caratheodory from scratch unless explicitly asked.

LLMwiki: [Stieltjes measures from distribution-like functions](#); ID: `durrett_ch1_stieltjes_measure`

Extension from Semialgebras

Proof template

A measure-like function on a semialgebra can be extended when it has the right countable additivity or continuity behavior on the generating class.

1. Start with simple pieces in $\mathcal{S}$.
2. Pass to finite disjoint unions of pieces.
3. Check countable additivity or continuity.
4. Extend to $\sigma(\mathcal{S})$.

LLMwiki: Extension from semialgebras; ID: durrett_ch1_semialgebra_extension

Multidimensional Distribution Functions

Theorem

A function F on $\mathbb{R}^d$ satisfying right-continuity, correct limits, and nonnegative rectangle increments defines a probability measure on $\mathcal{B}(\mathbb{R}^d)$.

$$\mu((a_1, b_1] \times \cdots \times (a_d, b_d]) = \Delta_{a,b} F \geq 0.$$

Pitfall

Coordinatewise monotonicity is not enough. Rectangle probabilities must be nonnegative.

LLMwiki: Multidimensional distribution functions; ID: durrett_ch1_product_distribution_function

How to Prove Generated Sigma-Field Claims

Standard two-inclusion proof

To show $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^d)$:

1. Show every set in $\mathcal{C}$ is Borel, so $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\mathbb{R}^d)$.
2. Show every open set belongs to $\sigma(\mathcal{C})$.
3. Use a countable rational base: every open set is a countable union of rational rectangles or balls.

LLMwiki: Sigma-field closure rules; ID: durrett_ch1_sigma_field

LLMwiki: Extension from semialgebras; ID: durrett_ch1_semialgebra_extension

Exercise 1: Countable/Co-countable Probability Space

Let $\Omega = \mathbb{R}$, let $\mathcal{F}$ be the collection of all subsets A such that A or A^c is countable, and define

$$\mathbb{P}(A) = \begin{cases} 0, & A \text{ is countable,} \\ 1, & A^c \text{ is countable.} \end{cases}$$

Show that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

LLMwiki: Countable/co-countable probability measure; ID: exercise_method_countable_cocountable_measure

Exercise 1: Proof Skeleton

1. $\emptyset, \mathbb{R} \in \mathcal{F}$, and complement closure is built into the definition.
2. For $A_n \in \mathcal{F}$, if all A_n are countable, then $\bigcup_n A_n$ is countable.
3. If some A_k is co-countable, then

$$\left(\bigcup_n A_n \right)^c \subseteq A_k^c,$$

so the union is co-countable.

4. For disjoint A_n , either all are countable and both sides of countable additivity are 0, or exactly one is co-countable and both sides are 1.

LLMviki: Countable/co-countable probability measure; ID: exercise_method_countable_cocountable_measure

Exercise 2: Borel Field from Half-Open Rectangles

Recall the definition of $\mathcal{S}_d$ from Example 1.1.5:

$$\mathcal{S}_d = \{\emptyset\} \cup \{(a_1, b_1] \times \cdots \times (a_d, b_d] : -\infty \leq a_i < b_i \leq \infty\}.$$

Show that $\sigma(\mathcal{S}_d) = \mathcal{B}(\mathbb{R}^d)$, the Borel subsets of $\mathbb{R}^d$.

LLMwiki: Borel sigma-field from half-open rectangles; ID:
`exercise_method_borel_generated_by_half_open_rectangles`

Exercise 2: Proof Skeleton

1. Since each half-open rectangle is Borel, $\sigma(\mathcal{S}_d) \subseteq \mathcal{B}(\mathbb{R}^d)$.
2. If $q_i < r_i$ are rational, then

$$(q_1, r_1) \times \cdots \times (q_d, r_d) = \bigcup_{m \geq 1} (q_1, r_1 - 1/m] \times \cdots \times (q_d, r_d - 1/m],$$

omitting empty terms.

3. Every open $G \subseteq \mathbb{R}^d$ is a countable union of rational open rectangles, hence $G \in \sigma(\mathcal{S}_d)$.
4. Therefore $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{S}_d)$.

LLMwiki: Borel sigma-field from half-open rectangles; ID:
`exercise_method_borel_generated_by_half_open_rectangles`

Exercise 3: Countably Generated Borel Field

A sigma-field $\mathcal{F}$ is said to be countably generated if there is a countable collection $\mathcal{C} \subseteq \mathcal{F}$ such that $\sigma(\mathcal{C}) = \mathcal{F}$. Show that $\mathcal{B}(\mathbb{R}^d)$ is countably generated.

LLMviki: Borel sigma-field from half-open rectangles; ID:
`exercise_method_borel_generated_by_half_open_rectangles`

Exercise 3: Proof Skeleton

Use the countable class

$$\mathcal{C} = \{B(q, r) : q \in \mathbb{Q}^d, r \in \mathbb{Q}, r > 0\}.$$

This is a countable base for the Euclidean topology. Since every open set is a countable union of balls from $\mathcal{C}$, all open sets lie in $\sigma(\mathcal{C})$. Thus $\mathcal{B}(\mathbb{R}^d) \subseteq \sigma(\mathcal{C})$. The reverse inclusion follows because every ball in $\mathcal{C}$ is open and hence Borel.

LLMwiki: Borel sigma-field from half-open rectangles; ID:
`exercise_method_borel_generated_by_half_open_rectangles`

Exercise 4: Increasing Union of Sigma-Fields

If $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots$ are sigma-algebras, then:

- (i) Show that $\bigcup_i \mathcal{F}_i$ is an algebra.
- (ii) Give an example to show that $\bigcup_i \mathcal{F}_i$ need not be a sigma-algebra.

LLMwiki: Increasing union of sigma-fields can fail countable closure; ID:
`exercise_method_increasing_sigma_fields_not_sigma_field`

Exercise 4: Counterexample

Let $\Omega = \mathbb{N}$ and

$$\mathcal{F}_n = \sigma(\{1\}, \{2\}, \dots, \{n\}).$$

Then $\bigcup_n \mathcal{F}_n$ contains every finite set. In particular, each singleton $\{2k\}$ belongs to $\bigcup_n \mathcal{F}_n$. But

$$E = \{2, 4, 6, \dots\} = \bigcup_{k \geq 1} \{2k\}$$

is neither finite nor cofinite relative to any finite initial-coordinate sigma-field, so $E \notin \bigcup_n \mathcal{F}_n$. Hence the union is not closed under countable unions.

LLMwiki: Increasing union of sigma-fields can fail countable closure; ID:
`exercise_method_increasing_sigma_fields_not_sigma_field`

Exercise 5: Asymptotic Density Is Not an Algebra

A set $A \subseteq \{1, 2, \dots\}$ is said to have asymptotic density θ if

$$\lim_{n \rightarrow \infty} \frac{|A \cap \{1, 2, \dots, n\}|}{n} = \theta.$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a sigma-algebra?
Is $\mathcal{A}$ an algebra?

LLMwiki: Block construction for asymptotic density counterexamples; ID:
`exercise_method_asymptotic_density_block_counterexample`

Exercise 5: Block Counterexample

Let $N_k = 2^{2^k}$ and $I_k = \{N_{k-1} + 1, \dots, N_k\}$. Let E be the even integers. Define

$$B \cap I_k = \begin{cases} E \cap I_k, & k \text{ odd,} \\ E^c \cap I_k, & k \text{ even.} \end{cases}$$

Then E has density $1/2$ and B has density $1/2$. But $E \cap B$ has density near $1/2$ at the end of odd blocks and near 0 at the end of even blocks. Since $N_k/N_{k-1} \rightarrow \infty$, the latest block dominates previous blocks, so the density of $E \cap B$ does not exist.

LLMwiki: [Block construction for asymptotic density counterexamples](#); ID:
`exercise_method_asymptotic_density_block_counterexample`

Checklist: How to Attack 1.1 Problems

1. Need to prove “probability space”? Check sigma-field, normalization, countable additivity.
2. Need to prove “generated sigma-field”? Use two inclusions and a countable rational base.
3. Need a limit of probabilities? Try continuity from below/above.
4. Need a counterexample to sigma-field closure? Use finite initial-coordinate sigma-fields.
5. Need a density-limit counterexample? Use rapidly growing blocks.

[LLMwiki: Chapter 1.1 exercise method cards](#)

Mini Practice

1. If $A_n \uparrow A$, prove $\mathbb{P}(A \setminus A_n) \rightarrow 0$.
2. Show $\sigma((a, b] : a < b) = \mathcal{B}(\mathbb{R})$.
3. Construct a finite algebra on $\Omega = \{1, 2, 3, 4\}$ generated by $\{1, 2\}$.
4. Suppose $A_n \downarrow \emptyset$. Explain why $\mathbb{P}(A_n) \rightarrow 0$.
5. Explain why an uncountable union of measurable sets need not be covered by the sigma-field axioms.